

# SAT Math Mastery · Trick Pack 1H

## SAT Trick Pack 1 — Part H

**Time** 75 minutes: 55 min solving + 20 min self-check and error log

**Topics** conditional probability · line and circle tangency · cofunction identities · perpendicular lines · intercept form · translations · regular polygons · Vieta and discriminant · rational formulas · composite solids · percent ratios · altitude relations · radical equations

**Name / Date** \_\_\_\_\_

### 1 · SKILLS & IDEAS

- S1.** Turn conditional-probability statements into linear equations in the unknown counts; the conditioned group is the denominator.
- S2.** Substitute a line into a circle or parabola and read the number of solutions from the discriminant.
- S3.** Use  $\sin \theta = \cos(90^\circ - \theta)$  and the second branch  $\sin A = \sin(180^\circ - A)$ ; filter by the allowed interval.
- S4.** Perpendicular lines in standard form:  $Ax + By = C$  has slope  $-\frac{A}{B}$ , so  $A_1A_2 + B_1B_2 = 0$ .
- S5.** Intercept form with one known zero: divide the two point equations to cancel the leading coefficient.
- S6.** Read a translated graph as an evaluation of the original function at a shifted input.
- S7.** Regular polygons: base angle  $\frac{180^\circ}{n}$ , exterior angle  $\frac{360^\circ}{n}$ ; match each given angle to the right triangle angle.
- S8.** Undo outside operations first, then inside, to recover points of an unknown linear function.
- S9.** Isolate a variable in a rational formula by one common denominator, or test with simple values.
- S10.** Composite solids: subtract the removed volume, convert cubic units with the cube of the length factor, multiply by density.
- S11.** Divide two percent equations that share a base to cancel it.
- S12.** Altitude to the hypotenuse:  $CD^2 = AD \cdot DB$ ,  $AC^2 = AD \cdot AB$ ,  $BC^2 = DB \cdot AB$ .
- S13.** Radical equations: square, then check every root against the sign condition on the other side.

### 2 · REQUIRED FACTS

- $P(A | B) = \frac{\text{count}(A \text{ and } B)}{\text{count}(B)}$ : the denominator is only the group named after “given”.
- A line meets a circle or parabola in 2, 1 or 0 points as the discriminant of the substituted quadratic is positive, zero or negative.
- $\sin \theta = \cos(90^\circ - \theta)$ ;  $\sin A = \sin B$  gives  $A = B + 360^\circ n$  or  $A = 180^\circ - B + 360^\circ n$ .
- Two non-vertical lines are perpendicular exactly when the product of their slopes is  $-1$ .
- $f(x - h) + k$  shifts the graph  $h$  right and  $k$  up; its  $y$ -intercept is  $f(-h) + k$ .
- Regular  $n$ -gon: interior angle  $180^\circ - \frac{360^\circ}{n}$ ; an isosceles triangle on two consecutive sides has base angles  $\frac{180^\circ}{n}$ .
- $1 \text{ cm}^3 = 1000 \text{ mm}^3$  (the factor is cubed); mass = density  $\times$  volume.
- A square-root equation  $\sqrt{u} = v$  requires  $v \geq 0$ ; squaring can create extraneous roots.
- For  $x^2 + px + q$  with real roots: sum  $= -p$ , product  $= q$ ; the roots are non-real exactly when  $p^2 - 4q < 0$ .

### 3 · REQUIRED FORMULAS

$$\Delta = b^2 - 4ac, \quad r + s = -\frac{b}{a}, \quad rs = \frac{c}{a}, \quad m = -\frac{A}{B}, \quad A_1A_2 + B_1B_2 = 0, \quad CD^2 = AD \cdot DB$$

$$\text{ext. angle} = \frac{360^\circ}{n}, \quad \text{base angle} = \frac{180^\circ}{n}, \quad V_{\text{cyl}} = \pi r^2 h, \quad \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}, \quad \sin \theta = \cos(90^\circ - \theta)$$

## 4 · MINI THEORY SHEET

**Conditional counts.** Write each probability as a fraction of its own group, cross-multiply, and solve the two linear equations. Never divide by the overall total unless the sentence says “of all”.

**Tangency and parameters.** Substitute, collect a quadratic, set  $\Delta = 0$  for one solution, and only then apply any sign condition on the parameter to pick the root.

**Trig equations.** Convert to one function, write both branches, add multiples of  $360^\circ$ , and keep only the values in the stated interval.

**Transformations.** Outside operations (multiply, add) act on outputs, inside operations act on inputs. To recover  $f$ , undo the outside first; a point  $(p, q)$  of  $g(x) = af(x - h) + k$  gives  $f(p - h) = \frac{q - k}{a}$ .

**Checking roots.** After squaring, test each root in the original equation, or impose the sign condition before counting.

**Speed shortcuts.**

| Situation                          | Shortcut                                                                                                             |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| “of those who ...”                 | the conditioned group is the denominator; cross-multiply the fractions                                               |
| line tangent to a circle           | $\Delta = 0$ right after substitution; then use the sign condition                                                   |
| $\sin(\cdot) = \cos(\cdot)$        | rewrite $\cos B = \sin(90^\circ - B)$ ; solve both branches; filter                                                  |
| perpendicular, standard form       | $A_1A_2 + B_1B_2 = 0$ ; no slopes needed                                                                             |
| one known zero, two points         | divide the two point equations to cancel $a$                                                                         |
| shift right $h$                    | new $y$ -intercept is $f(-h)$ (not $f(h)$ )                                                                          |
| regular polygon, diagonal          | isosceles triangle: base $\frac{180^\circ}{n}$ ; apex = interior angle                                               |
| transformed linear graph           | undo outside first, then inside; slope from two recovered points                                                     |
| rational formula                   | plug simple values for the other letters and test the options                                                        |
| drilled solid and density          | factor common $\pi h$ , convert with the cube of the length factor                                                   |
| two percent equations, same base   | divide: the base cancels                                                                                             |
| altitude to the hypotenuse         | $\text{leg}^2 = \text{hypotenuse} \times \text{adjacent segment}$ ; $\text{altitude}^2 = \text{product of segments}$ |
| $\sqrt{u} = v$ with a parameter    | square, then require $v \geq 0$ for each root                                                                        |
| quadratic factor with no real zero | strict $\Delta < 0$ ; test the boundary value before counting it                                                     |

## 5 · GUIDED EXAMPLES

## Example 1 — Conditional probability to a linear system

A club has 50 members,  $j$  juniors and  $s$  seniors. Half the juniors and  $\frac{1}{5}$  of the seniors are in the band. A random band member is a junior with probability  $\frac{5}{8}$ . Find  $j$ .

$$\frac{j/2}{j/2 + s/5} = \frac{5}{8} \Rightarrow 8 \cdot \frac{j}{2} = 5 \left( \frac{j}{2} + \frac{s}{5} \right) \Rightarrow 4j = \frac{5j}{2} + s \Rightarrow s = \frac{3j}{2}.$$

$$j + s = 50 \Rightarrow \frac{5j}{2} = 50 \Rightarrow j = 20. \text{ (Check: band} = 10 + 6 = 16, \text{ juniors } \frac{10}{16} = \frac{5}{8}.)$$

## Example 2 — Tangent line: discriminant first, sign second

The line  $y = x + c$ ,  $c > 0$ , meets  $x^2 + y^2 = 8$  in exactly one point. Find  $c$ .

$$x^2 + (x + c)^2 = 8 \Rightarrow 2x^2 + 2cx + c^2 - 8 = 0. \Delta = 4c^2 - 8(c^2 - 8) = 64 - 4c^2 = 0 \Rightarrow c^2 = 16.$$

$c = \pm 4$ ; the condition  $c > 0$  gives  $c = 4$ .

## Example 3 — Radical equation: square, then test the sign

Solve  $\sqrt{x+6} = x$ .

Square:  $x + 6 = x^2 \Rightarrow x^2 - x - 6 = 0 \Rightarrow x = 3$  or  $x = -2$ .

The right side must be  $\geq 0$ , so  $x = -2$  is extraneous ( $\sqrt{4} = 2 \neq -2$ ). Only  $x = 3$  works.

**6 · PRACTICE SET — 16 PROBLEMS · ALL SAT CHALLENGE**

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

**1.** SAT CHALLENGE · NO CALCULATOR NEEDED

A survey of 240 adults asked whether they commute by bicycle and whether they own a helmet. Of the adults who commute by bicycle,  $\frac{3}{5}$  own a helmet. Of the adults who do not commute by bicycle,  $\frac{1}{4}$  own a helmet. If one adult is selected at random from those who own a helmet, the probability that the selected adult commutes by bicycle is  $\frac{12}{19}$ . How many of the 240 adults own a helmet?

(A) 60

(B) 95

(C) 144

(D) 100

**2.**

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 = 45$$

$$y = mx + 15$$

In the system above,  $m$  is a negative constant. If the system has exactly one solution  $(x, y)$ , what is the value of  $x$ ?

(A)  $-3$ (B)  $3$ (C)  $6$ (D)  $-6$ **3.**

SAT CHALLENGE · CALCULATOR

For  $0 < x < 45$ , the equation

$$\sin((5x - 20)^\circ) = \cos((x + 14)^\circ)$$

has more than one solution. What is the sum of all solutions  $x$  in this interval?

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the  $xy$ -plane, line  $\ell$  has equation  $5x - 2y = 11$  and line  $m$  has equation  $ax + by = d$ , where  $a$ ,  $b$  and  $d$  are positive constants and  $a + b = 21$ . Lines  $\ell$  and  $m$  are perpendicular, and line  $m$  passes through the point  $(4, 3)$ . What is the value of  $d$ ?

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function  $f$  has a zero at  $x = 2$ . The graph of  $y = f(x)$  in the  $xy$ -plane passes through the points  $(0, -16)$  and  $(5, 54)$ . What is the minimum value of  $f$ ?

(A)  $-18$ (B)  $-16$ (C)  $-9$ (D)  $-1$

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of the quadratic function  $f$  in the  $xy$ -plane passes through the points  $(0, 1)$ ,  $(1, -2)$  and  $(3, 4)$ . The graph of  $y = g(x)$  is obtained by translating the graph of  $y = f(x)$  3 units to the right and 5 units down. What is the  $y$ -coordinate of the  $y$ -intercept of the graph of  $g$ ?

(A) 34

(B) 39

(C)  $-1$ 

(D) 29

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

Points  $A$ ,  $B$ ,  $C$  and  $D$  are consecutive vertices of a regular polygon with  $n$  sides, and  $\overline{AC}$  is a diagonal. The measure of  $\angle BAC$  is  $(x + 3)^\circ$  and the measure of  $\angle ACD$  is  $(10x + 15)^\circ$ , where  $x$  is a constant. What is the value of  $n$ ?

(A) 24

(B) 8

(C) 12

(D) 15

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function  $f$  is linear. The function  $g$  is defined by  $g(x) = -2f(x - 1) + 5$ , and the graph of  $y = g(x)$  passes through the points  $(0, 3)$  and  $(2, -5)$ . The graph of  $y = h(x)$  is the graph of  $y = f(x)$  translated 2 units to the left and 6 units up. What is the  $y$ -coordinate of the  $y$ -intercept of the graph of  $h$ ?

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9.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$y = 3x^2 + 4$$

$$y = 1 - kx$$

In the system above,  $k$  is a constant. For which of the following values of  $k$  does the system have two distinct solutions  $(x, y)$  whose  $x$ -coordinates are both negative?

(A) 6

(B) 7

(C)  $-7$ 

(D) 2

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10.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{2}{a} - \frac{3}{b} = \frac{5}{c}$$

In the equation above,  $a$ ,  $b$  and  $c$  are nonzero constants and  $2c \neq 5a$ . Which of the following expresses  $b$  in terms of  $a$  and  $c$ ?

(A)  $b = \frac{3ac}{5a - 2c}$

(B)  $b = \frac{2c - 5a}{3ac}$

(C)  $b = \frac{3ac}{2c + 5a}$

(D)  $b = \frac{3ac}{2c - 5a}$

11.

SAT CHALLENGE · CALCULATOR

A brass bushing is a solid right circular cylinder with diameter 60 millimeters and height 40 millimeters. A cylindrical hole of diameter 20 millimeters is drilled into one flat face, along the axis, to a depth of 25 millimeters. The density of brass is 7.8 grams per cubic centimeter. Which of the following is closest to the mass of the bushing, in grams? (1 cm = 10 mm)

(A) 821

(B) 882

(C) 82,090

(D) 784

12.

SAT CHALLENGE · CALCULATOR

A retailer's total sales were the same in March and April. In March, online sales were  $p$  percent of the total sales and came to \$18,000. In April, online sales were  $(p + 8)$  percent of the total sales and came to \$25,200. What were the retailer's total sales, in dollars, for one of these months?

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13.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle  $ABC$ , the angle at  $C$  is a right angle. Point  $D$  lies on  $\overline{AB}$  so that  $\overline{CD} \perp \overline{AB}$ . The length of  $\overline{CD}$  is 6, and  $DB$  is 5 units longer than  $AD$ . What is the value of  $\sin B$ ?

- (A)  $\frac{6}{13}$  (B)  $\frac{2\sqrt{13}}{13}$   
(C)  $\frac{3\sqrt{13}}{13}$  (D)  $\frac{4}{13}$

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**14.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The function  $f$  is linear. The function  $g$  is defined by  $g(x) = \frac{f(x+2) - x}{x-3}$  for  $x \neq 3$ . Some values of  $g$  are shown in the table below.

|        |   |     |
|--------|---|-----|
| $x$    | 1 | 5   |
| $g(x)$ | 3 | -11 |

What is the  $y$ -intercept of the graph of  $y = f(x)$  in the  $xy$ -plane?

(A) -5

(B) -4

(C) 4

(D) -3

**15.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For how many integer values of  $k$  with  $-10 \leq k \leq 10$  does the equation

$$\sqrt{x+k} = x-2$$

have exactly one real solution?

**16.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial  $p(x) = x^3 + bx^2 + cx + 12$  has integer coefficients  $b$  and  $c$ , and  $x + 3$  is a factor of  $p(x)$ . If  $p$  has exactly one real zero, what is the greatest possible value of  $c$ ?

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